# BIPASHA BANERJEE

## CONTACT INFORMATION

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**GOOGLE SCHOLAR:** https://scholar.google.com/citations?user=B8WSVAEAAAAJ&hl=en

## EDUCATION

**Ph.D. COMPUTER SCIENCE**, VIRGINIA TECH, BLACKSBURG, VA 2024
**Dissertation**: Banerjee, B. (2024). Improving Access to ETD Elements Through Chapter Categorization and Summarization, Ph.D. Thesis, Virginia Tech, 2024. https://hdl.handle.net/10919/120890

**M.S. COMPUTER SCIENCE,** VIRGINIA TECH, BLACKSBURG, VA 2022

**B.Tech. COMPUTER SCIENCE & ENGINEERING**
WEST BENGAL UNIVERSITY OF TECHNOLOGY, KOLKATA, INDIA 2015

## PROFESSIONAL EXPERIENCE

**ASSISTANT PROFESSOR,**
**AI RESEARCH SCIENTIST, DIGITAL LIBRARIES**
August 2024 - Present
University Libraries, Virginia Tech, Blacksburg, VA

**GRADUATE RESEARCH ASSISTANT**
August 2019 – August 2024
University Libraries, Virginia Tech, Blacksburg, VA

**GLOBAL PRODUCT AND TECHNOLOGY INTERN**
May – August 2019
ADP, Alpharetta, GA

**GRADUATE TEACHING ASSISTANT**
January – May 2019
Computer Science, Virginia Tech, Blacksburg, VA

**ASSISTANT SYSTEMS ENGINEER - FULL TIME**
April 2016 – November 2017
TATA Consultancy Services, Kolkata, India

## RESEARCH INTERESTS

I am an Assistant Professor at Virginia Tech, where I currently work as an AI Research Scientist, Digital Libraries for the University Libraries. My research mainly focuses on digital libraries, natural language processing, machine learning, artificial intelligence, and scholarly big data. I lead a team of extraordinary students working exciting research such as information extraction from multi-modal digital library collections, detecting relevance of literature based in a systematic review framework, automatic detection of walkability indicators in retirement communities.

## RECENT PUBLICATIONS & TALKS

1. Banerjee, B., Goyne, J., & Bohra, B. (2026, October). AI-Assisted Workflows for Historical Map Metadata Extraction. *In Proceedings of the 26th ACM/IEEE Joint Conference on Digital Libraries.* DOI: https://doi.org/10.1145/3805696.3846471

2. Kang, M., Banerjee, B., & Comer, C. C. (2026, October). Evaluating and explaining LLM classification for automatic screening in Systematic Reviews. *In Proceedings of the 26th ACM/IEEE Joint Conference on Digital Libraries Article No.:* 61 DOI: https://doi.org/10.1145/3805696.3846474
3. German, F., Benerjee, B., Keith-Norambuena, B., Ingram, W., Fox, E., & North, C. (2026, October). Structuring The Top-K Results: The Case for Narrative Extraction, Structuring, and Visualization in Scholarly Research Tools. In *Proceedings of the 26th ACM/IEEE Joint Conference on Digital Libraries.* DOI: https://doi.org/10.1145/3805696.3846480

4. Salsabil, L., Obadage, R. R., Banerjee, B., Alam, S., Ingram, W., Fox, E., & Wu, J. (2026, October). End-to-End Contextualized URL Extraction from Scholarly Papers. In *Proceedings of the 26th ACM/IEEE Joint Conference on Digital Libraries* DOI: https://doi.org/10.1145/3805696.3846039

5. Obadage, R. R., Salsabil, L., Alam, S., Ingram, W., Banerjee, B., Fox, E., & Wu, J. (2026, October). URL Extraction from Scholarly Documents: A Cross-Format Comparative Analysis. In *Proceedings of the 26th ACM/IEEE Joint Conference on Digital Libraries* DOI: https://doi.org/10.1145/3805696.3846023

6. Ingram, W. A., Peloquin, S., Dommanige, C., Salsabil, L., Obadage, R. R., Banerjee, B., ... & Fox, E. A. (2026, October). A Longitudinal Panel of Open Access Data and Software URLs from PLOS Publications. In *Proceedings of the 26th ACM/IEEE Joint Conference on Digital Libraries* DOI: https://doi.org/10.1145/3805696.384848

7. Banerjee, B. (2026). Digital Libraries in the Age of AI: Opportunities and Challenges. *ACM SIGWEB Newsletter*, *2026*(Summer), DOI: https://doi.org/10.1145/3830455.3830457

8. Salsabil. L., Obadage, R. R., Banerjee, B., Abeysinghe Y., Alam, S., Färber, M., Ingram, W. A., Fox, E.A. & Wu, J. Context-Based URL Classification for Open Access Datasets and Software in Scholarly Document. In 2025, The 25th ACM/IEEE Join Conference on Digital Libraries (JCDL) DOI: 10.1109/JCDL67857.2025.00031

9. Ingram, W. A, Banerjee, B. & Fox, E.A. Learning from LLM Disagreement in Retrieval Evaluation. In 2025, The 25th ACM/IEEE Join Conference on Digital Libraries (JCDL). DOI: 10.1109/JCDL67857.2025.00024

10. Kalir, H., German, F., Aboelnaga, Amr A., Banerjee, B., Eldardiry, H. & Ingram W. A. Retrieval-Augmented LLMs for ETD Subject Classification. In 2025 IEEE International Conference on Big Data (BigData). 10.1109/BigData66926.2025.11402222

11. Jadhav, A., Banerjee, B., & Goyne J. Using an Ensemble Approach for Layout Detection and Extraction from Historical Newspapers. In 2025 IEEE International Conference on Big Data (BigData). 10.1109/BigData66926.2025.11401834

12. Miller, C., & Banerjee, B. Optical Character Recognition for Pre-Digital Historical Documents using Large Language Models. In 2025, The 29th International Conference on Machine Learning and Applications. 10.1109/ICMLA66185.2025.00025

13. Klair, H., German, F., Banerjee, B., & Ingram, W. A. (Accepted). Evaluating Human-LLM Alignment in ETD Subject Classification. In 2025, The 29th International Conference on Theory and Practice of Digital Libraries. https://link.springer.com/chapter/10.1007/978-3-032-06136-2_6

14. Ingram, W. A., Banerjee, B., & Fox, E. A. (2025). When LLMs Disagree: Diagnosing Relevance Filtering Bias and Retrieval Divergence in SDG Search. LLM4Eval at SIGIR 2025 DOI: https://doi.org/10.48550/arXiv.2507.02139

15. Banerjee, B., Goyne, J., & Ingram, W. A. (2024, December). Making History Readable. In 2024 IEEE International Conference on Big Data (BigData) (pp. 8620-8622). IEEE. DOI: 10.1109/BigData62323.2024.10826028

16. Banerjee, B., Ingram, W. A., & Fox, E. A. (2024, December). Automating Chapter-Level Classification for Electronic Theses and Dissertations. In 2024 IEEE International Conference on Big Data (BigData) (pp. 2400-2409). IEEE. DOI: 10.1109/BigData62323.2024.10825418

17. Ingram, W. A., Banerjee, B., & Fox, E. A. (2024, December). Agentic AI for Improving Precision in Identifying Contributions to Sustainable Development Goals. In 2024 IEEE International Conference on Big Data (BigData) (pp. 8677-8679). IEEE. DOI: 10.1109/BigData62323.2024.10825072

18. Ingram, W. A., Wu, J., Kahu, S. Y., Manzoor, J. A., Banerjee, B., Ahuja, A., ... & Fox, E. A. (2024). Building datasets to support information extraction and structure parsing from electronic theses and dissertations. International Journal on Digital Libraries, 25(2), 175-196. DOI: https://doi.org/10.1007/s00799-024-00395-4

19. Park, S. H., Banerjee, B., Ingram, W. A., & Fox, E. A. (2023). Case Study of Analyzing the Variety of ETD Layouts. URL: https://ir.inflibnet.ac.in/server/api/core/bitstreams/27afc90c-1a82-470e-8baa-5c1a3769426b/content

20. Chekuri, S., Chandrasekar, P., Banerjee, B., Park, S. H., Masrourisaadat, N., Ahuja, A., ... & Fox, E. A. (2023, June). Integrated digital library system for long documents and their elements. In 2023 ACM/IEEE Joint Conference on Digital Libraries (JCDL) (pp. 13-24). IEEE. DOI: 10.1109/JCDL57899.2023.00012

21. Banerjee, B., Ingram, W. A., Wu, J., & Fox, E. A. (2022, December). Applications of data analysis on scholarly long documents. In 2022 IEEE International Conference on Big Data (Big Data) (pp. 2473-2481). IEEE. DOI: 10.1109/BigData55660.2022.10020935

22. Banerjee, B., Jude, P.M., Ingram, W.A., Luther K., Fox, E.A. (2022). Help Me Help You - A Mixed-Initiative Approach To Explore Book-length Documents. Workshop on Human-in-the-loop Data Curation at ACM CIKM 2022 https://hilworkshops.github.io/hil-dc2022/

23. Banerjee, B. (2022, June). Opening scholarly documents through text analytics. In Proceedings of the 22nd ACM/IEEE Joint Conference on Digital Libraries (pp. 1-2). DOI: https://doi.org/10.1145/3529372.3530948

24. Uddin, S., Banerjee, B., Wu, J., Ingram, W. A., & Fox, E. A. (2021, December). Building A large collection of multi-domain electronic theses and dissertations. In 2021 IEEE International Conference on Big Data (Big Data) (pp. 6043-6045). IEEE. DOI: 10.1109/BigData52589.2021.9672058

25. Banerjee, B., William A. Ingram, Jian Wu, and Edward A. Fox (2021), Applications of Mining ETDs. ETD 2021 Conference: Open scholarship in a post-pandemic world. DOI: https://doi.org/10.26226/morressier.614c9b8c87a68d83cb5d59b2

26. Aromando, J., Banerjee, B., Ingram, W. A., Jude, P., & Kahu, S. (2020). Classification and extraction of information from ETD documents. In Fox, E.A. course CS 6604: Digital Libraries, 2019. http://hdl.handle.net/10919/96645

27. Khawas, P., Banerjee, B., Kim, Y., Zhao, S., & Fan, Y. (2018). Big Data Text Summarization for Maryland Shooting Collection. In Fox, E.A. course CS 5984: Big Data Text Summarization 2018. http://hdl.handle.net/10919/86407

28. Ingram, W. A., Banerjee, B., & Fox, E. A. (2019). Summarizing ETDs with deep learning. Cadernos BAD, 1, 46-52. https://opening-etds.github.io/files/2014-5500-1-PB.pdf

## INVITED TALKS

1. Banerjee, B. (2026). Invited speaker at the Library of Congress AI Literacy and Learning Summit. Washington DC.

2. Miller, C., Banerjee, B. (2025). Preserving and Understanding the Past through Large Language Models. Generative AI in Libraries Conference. Recording available.

3. Banerjee, B., Goyne, J. (2025). AI-driven Solutions for Digital Library Collections. Generative AI in Libraries Conference. Recording available.

## RECENT GRANTS

1. "Library-wide, Machine-Generated Metadata as First-Class Structural Components". PI: Bipasha Banerjee, Co-PI: Jennifer L. Goyne, William A. Ingram. A three-year applied research grant from Institute of Museum and Library Services. Awarded $473,403

2. "Auditable AI Workflows for Evidence Synthesis Library Services". PI: Bipasha Banerjee, Co-PI: C. Cozette Comer and William A. Ingram. Two-year planning grant from Institute of Museum and Library Services. Awarded $178,994.

3. "Resilient Post-Industrial Landscapes along the James River: A Collaborative AI and Archival Approach" PI: Dr. Irene Curulli, School of Design, Co-PI: Dr. Bipasha Banerjee. University Libraries – Awarded $10,000

4. "Human-Centered GeoAI for Age-Friendly Walkability at Warm Hearth Village" PI: Dr. Eunju Hwang, School of Design, Co-PIs: Dr. Bipasha Banerjee, Digital Library, Dr. Junghwan Kim, Geography. Co-I: Jennifer Goyne, Digital Library, Sara McCarter, WHV Community Partner, Dr. Jaehwan Kim, Real Estate, Kongju National University, South Korea - Awarded $10,000

## PATENTS

Co-inventor on: US Patent Application 18/585,685; filed February 23, 2024; published 8/29/2024; Document ID US 20240289356 A1: Structured Documentation Access for Electronic Documents: Edward A. Fox, William A. Ingram, Aman Ahuja, Satvik Chekuri, and Bipasha Banerjee. https://patents.google.com/patent/US20240289356A1

## TEACHING EXPERIENCE

**FACULTY MENTOR** Virginia Tech, Blacksburg, VA, 2024 - Present
Advising graduate and undergraduate students on their research and curriculum

**GRADUATE STUDENT MENTOR** Virginia Tech, Blacksburg, VA, 2019-2024

**GRADUATE TEACHING ASSISTANT** Virginia Tech, Blacksburg, VA, JAN – MAY 2019

## PROFESSIONAL INVITED SERVICE

- Invited long paper reviewer for ACM TheWebConf 2027, September 2026
- Invited reviewer for Information and Culture, University of Texas Austin Press, June 2026
- Co-organized, developed and presented a pre-conference workshop titled "AI in Evidence Synthesis" at USAIN (US Ag Info Network) on May 2026.
- Co-chair, Conference Proceedings, IEEE/ACM Joint Conference on Digital Libraries, 2025, 2026.
- Invited reviewer for Springer Nature Scientometrics Journal Article, 2025.
- Invited as Program Committee Reviewer for IEEE/ACM Joint Conference on Digital Libraries 2022, 2023, 2024, 2025, 2026
- Co-organized a workshop titled "Hands-on digitizing texts with machine learning and AI". A three-hour long workshop open to VT community on using machine learning and LLMs for optical character recognition

## PROFESSIONAL MEMBERSHIPS

- Association for Computing Machinery (ACM) Member- #3768609
- Institute of Electrical and Electronics Engineers (IEEE) Member- #96669088

## PROFESSIONAL UNIVERSITY ACTIVITIES

- Selected to serve as a member of Library Continued Appointment Track Faculty Committee 2026-2028.
- Collaboration on a project geared toward using LLMs for University-wide analysis in partnership with University Libraries and the Office of Analytics & Institutional Effectiveness, Virginia Tech.
- Led breakout sessions at the workshop titled "Ensuring Scholarly Access to Government Archives and Records," organized by Virginia Tech University Libraries in partnership with Virginia Tech Center for Humanities and the U.S. National Archives and Records Administration (funded by The Andrew W. Mellon Foundation)
- Digital Library Research Laboratory student representative for CS grad council at Virginia Tech (2020-2023).

- Association of Women in Computing Member, Virginia Tech Chapter.
- Volunteered at VT CS grad recruiting weekend and career fair.

## AWARDS

- Recognized by Virginia Tech as a "National Distinction/Outstanding Contributors Program" faculty for FY 2025-26.
- Nominated and inducted into Sigma Xi Scientific Honor Society - 2025
- Recognized as an inventor by Virginia Tech for "identifying and disclosing a technology for FY 23" in 2024.
- Nominated for best student paper for "Integrated Digital Library System for Long Documents and their Elements" at IEEE/ACM JCDL 2023.
- Awarded a scholarship to attend the Grace Hopper Conference by the CS department at Virginia Tech in 2021 and the Association of Women in Computing (AWC, Virginia Tech Chapter) in 2022.
- Awarded free attendance at the International Conference in Pattern Recognition (ICPR), virtual 2020.
- "Student of the Year" awarded by the Institute of Engineering and Management for academics as well as contribution to the college in 2014.